SEIZURE LOCALIZATION USING QUANTUM ALGORITHMS

Final Project Evaluation

Group 12

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- Literature Survey
- Problem definition
- Objectives
- Proposed Method
- Architecture Diagram
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- Software & Hardware Requirements
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INTRODUCTION

- A seizure is a sudden burst of abnormal electrical activity in the brain that temporarily disrupts normal neuronal communication and can affect movement, awareness, or behavior.

- Epilepsy is a chronic neurological disorder in which a person experiences recurrent, unprovoked seizures due to abnormal brain activity.

- Epilepsy affects approximately 50 million people worldwide, making it one of the most common neurological disorders.

PURPOSE & NEED

- Seizures occur suddenly and unpredictably, posing serious risks to patient safety.
- Subtle changes in EEG signals appear before seizure onset but are often hidden within noisy data.
- The transition from normal brain activity to seizure is gradual, making the exact onset difficult to identify.
- Early detection of these changes can enable seizure prediction and timely intervention.

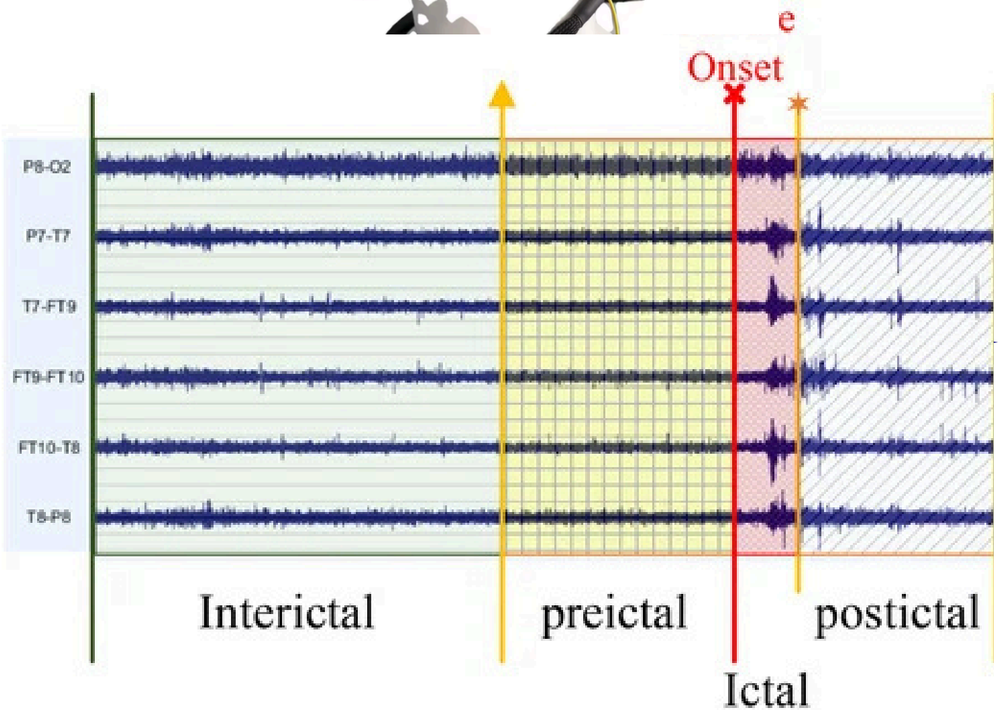
EEG AND SEIZURE STAGES

- Electroencephalography (EEG) is a primary clinical tool used to record electrical activity of the brain.

- EEG signals represent the collective electrical activity of neurons and allow abnormal brain activity associated with seizures to be observed.

- Seizure related brain activity progresses through multiple stages including interictal, preictal, ictal, and postictal phases.





LITERATURE SURVEY

Paper Title	Methodology	Inference
Seizure Detection using EEG: A Survey of Different Techniques (*S. Ahmad, R. A. Rafiq, and S. Rehman*) [1]	Reviewed EEG-based seizure detection pipeline: acquisition, preprocessing (filtering, artifact removal), feature extraction (time, frequency), and classification methods.	Highlighted that combining time- and frequency-domain features improves detection performance; noted lack of real-time predictive focus.
A Review of Feature Extraction for EEG Epileptic Seizure Detection and Classification (*S. Roy, M. Asif, and D. K. Vishwakarma*) [2]	Surveyed feature extraction methods: statistical (mean, variance, Hjorth), spectral (FFT, PSD), and time–frequency (wavelet). Compared traditional approaches with deep learning-based feature learning.	Concluded that handcrafted + deep learning features yield best results; emphasized wavelet-based methods for seizure transition analysis.
Comprehensive Review of EEG-Based Seizure Detection Techniques *(T. S. Alotaiby, S. A. Alshebeili and F. M. Alotaibi*) [3]	Summarized traditional and modern seizure detection approaches, including time, frequency, and time–frequency feature extraction; also considered seizure phases (preictal, ictal, postictal).	Showed that integrating multi-domain features enhances robustness; highlighted the importance of preictal detection for early intervention.

LITERATURE SURVEY

Paper Title	Methodology	Inference
EEG-Based Seizure Prediction via Model Uncertainty Learning (*Chang Li, Zhiwei Deng, Rencheng Song, Xiang Liu*) [4]	Proposed RepNet with Modified Monte Carlo Dropout (MMCD) for uncertainty-aware seizure prediction. Uses structural reparameterization to make the network lightweight and efficient.	Achieves high prediction accuracy with uncertainty estimation, enabling reliable clinical decision-making.
Monte Carlo Dropout for Uncertainty Estimation (*Gal & Ghahramani*) [5]	Dropout at inference approximates Bayesian uncertainty with multiple forward passes.	Provides uncertainty quantification, but increases inference time.
RepNet: Lightweight CNN with Structural Reparameterization (*Ding et al.) [6]*	Multi-branch CNN trained and converted to single-branch at deployment for low latency.	Well-suited for real-time inference, forming the backbone of the proposed model.

LITERATURE SURVEY

Paper Title	Methodology	Inference
Quantum Computing: Circuits, Algorithms, and Applications *(M A Shafique, Arslan Munir , Imran Latif) [7]*	Study of quantum computing fundamentals including qubits, superposition, entanglement, and circuits that is applicable to various fields.	Quantum computing principles like superposition and entanglement enable superior processing for complex tasks.
Quantum data encoding: comparative analysis of classical-to-quantum mapping techniques and their impact on machine learning accuracy *(Minati Rath and Hema Date) [8]*	A study on basis, angle, and amplitude encoding techniques for classical data and assesses performance on classical machine learning models.	Quantum data encoding can potentially enhance the accuracy of machine learning models. The choice of encoding technique can significantly impact the performance
QEEGNet: Quantum Machine Learning for Enhanced EEG Encoding. *(Chi-Sheng Chen, Samuel Yen-Chi Chen, Aidan Hung-Wen Tsai, Chun-Shu Wei) [9]*	Hybrid Quantum Classical architecture for EEG data encoding for improved analysis and classification.	Successful application of a hybrid quantum-classical model to EEG data provides a compelling proof of concept

LITERATURE SURVEY

Paper Title	Methodology	Inference
Quantum Machine-Based Decision Support System for the Detection of Schizophrenia from EEG Records *(Gamzepelin Aksoy Grégoire Cattan Subrata Chakraborty Murat Karabatak) [10]*	EEG data from 45 schizophrenic and 39 healthy adolescents were processed with DWT-based feature extraction and PCA, then classified using both classical ML models and a QSVM on the IBM Quantum simulator.	QSVM with Pauli X/Z maps achieved 100% accuracy, showing quantum ML's effectiveness for EEG classification, though performance depends heavily on the chosen feature map.
Heart Disease Detection using Quantum Computing and Partitioned Random Forest Methods *(Hanif Heidari, Gerhard Hellstern, Murugappan) [11]*	Two open-source heart disease datasets (Cleveland, Statlog) were used to test HQRF (Hybrid Quantum Random Forest), which partitions features into subsets, processes each with a Hybrid Quantum Neural Network, and aggregates outputs for prediction. Performance was compared against HQNN and classical models.	HQRF outperforms HQNN on small datasets, is more robust to outliers, leverages partitioning to overcome quantum hardware limits, and achieved 91.64% AUC on Statlog.

GAPS IDENTIFIED

- Limited exploration of advanced feature representations beyond traditional time- and frequency-domain approaches.
- Quantum and hybrid learning methods have been explored in other biomedical domains but remain minimally applied to EEG-based seizure prediction.
- Many approaches operate on fixed EEG segments, limiting the ability to capture temporal dependencies across longer sequences.
- Most existing studies focus on seizure detection rather than early prediction or warning systems.

PROBLEM DEFINITION

Seizure-related brain activity as it progresses through multiple stages, often marked by complex and inconsistent transitions. These irregular patterns make time-based localisation of seizures a significant challenge. This study aims to explore potential solutions to this challenge using quantum-based approaches.

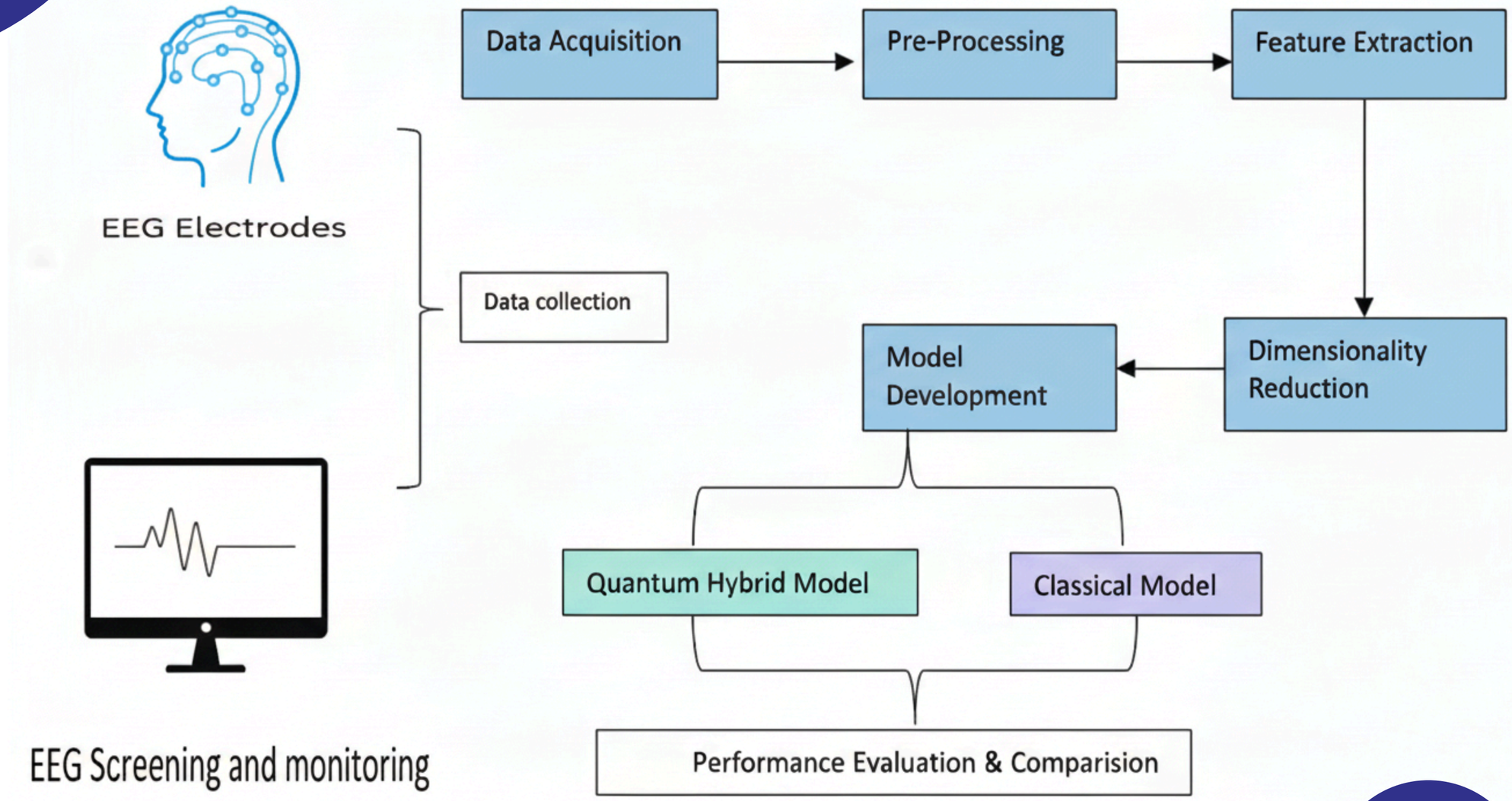
OBJECTIVES

- To study the EEG activity across different stages of seizure.
- To identify and characterise key EEG features associated with these stages.
- To develop computational models based on the extracted features for seizure localisation over time.
- To explore the application of quantum-based approaches to enhance modelling of complex EEG data towards an early warning system.
- To evaluate the effectiveness and limitations of the proposed methods in capturing seizure dynamics.

PROPOSED METHOD

- Segment EEG recordings into fixed-length labeled windows.
- Extract and Analyse EEG features from each window, including time-domain, frequency-domain, and quantum-inspired features.
- Train classical and hybrid quantum–classical models to study abnormal seizure states.
- Evaluate the effectiveness and feasibility of these models for preictal and ictal stage classification across temporal windows.
- Extend analysis toward preictal detection for early warning.

ARCHITECTURE DIAGRAM



MODULES

Module 1 : Data Preprocessing

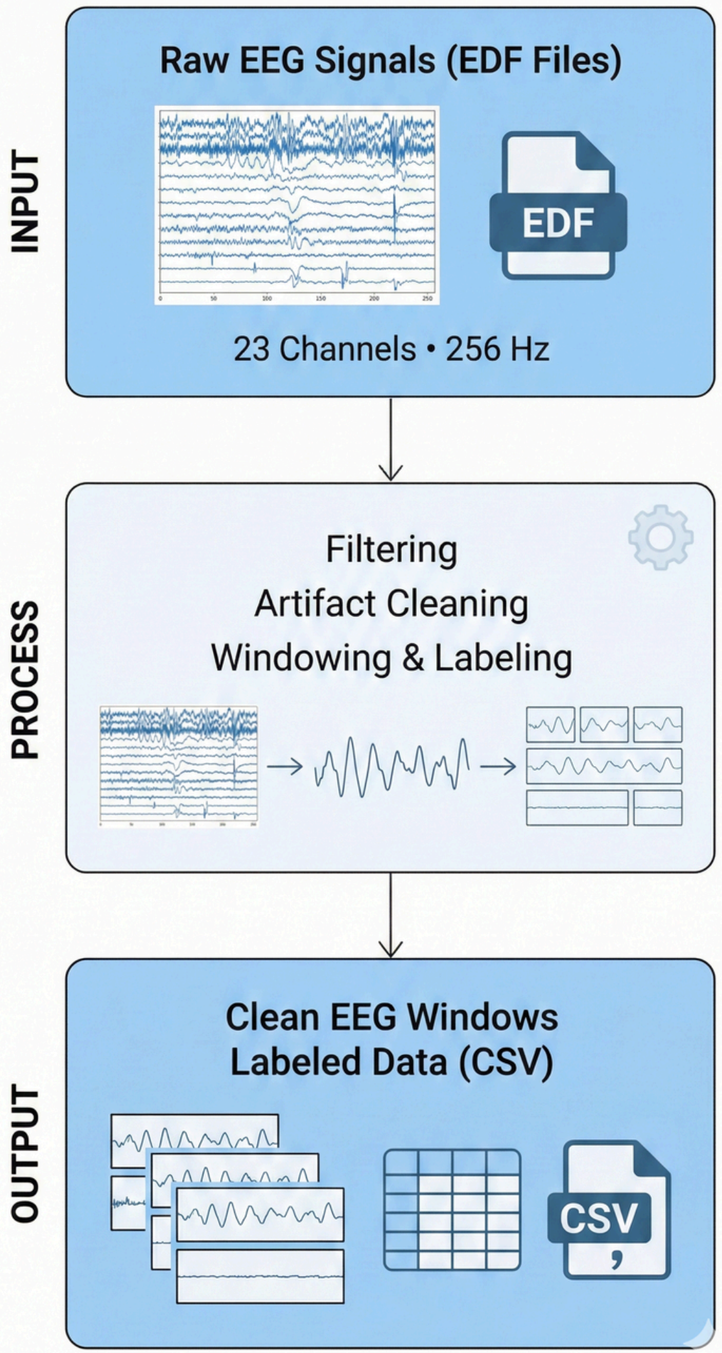
Module 2: Feature Extraction and Analysis

Module 3: Classical Model Development

Module 4: Hybrid Quantum Classical Model Development

Module 5: Model Evaluation and Comparison

MODULE 1
DATA PRE-PROCESSING



MODULE 1: DATA PRE-PROCESSING

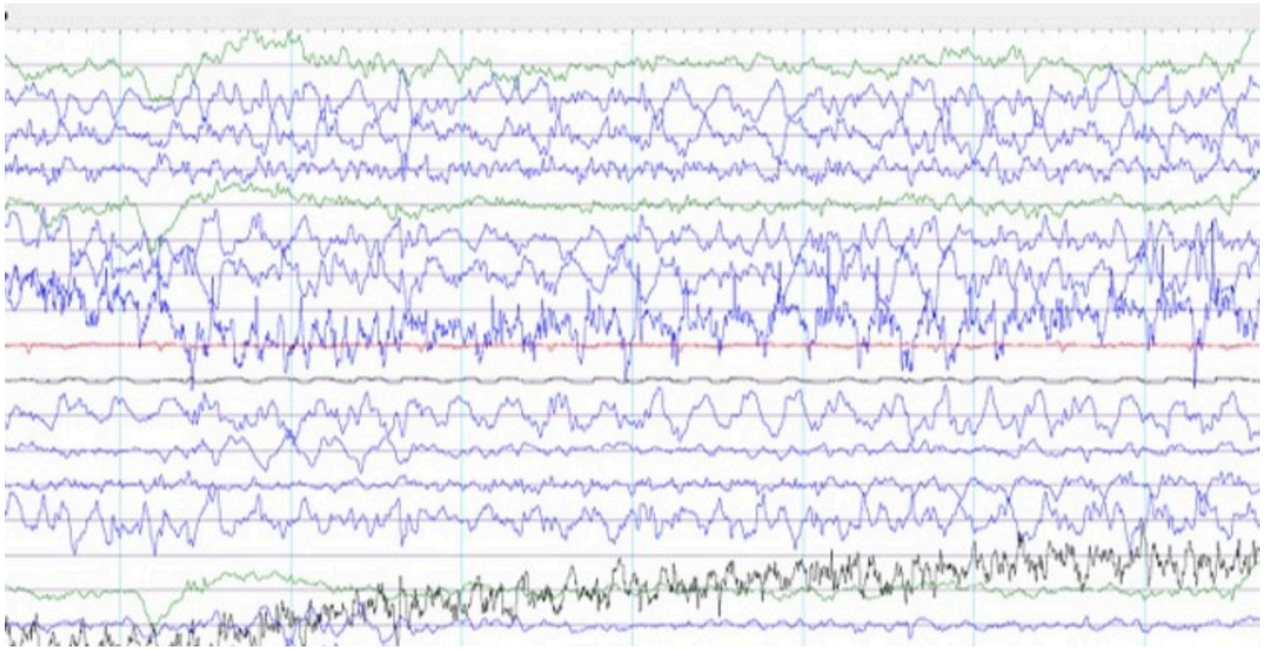
Datasets

CHB-MIT

- Dataset collected from Children’s Hospital Boston.
- Contains EEG recordings of 22 pediatric patients (5 males, 17 females; ages 1.5–22) with focal seizures.
- Recordings were obtained over several days following withdrawal of anti-seizure medication..
- Includes 23 EEG channels/electrode.
- Sampled at 256 Hz.
- Provided in EDF (European Data Format) files.

MODULE 1: DATA PRE-PROCESSING

CHB-MIT



chb12-summary.txt

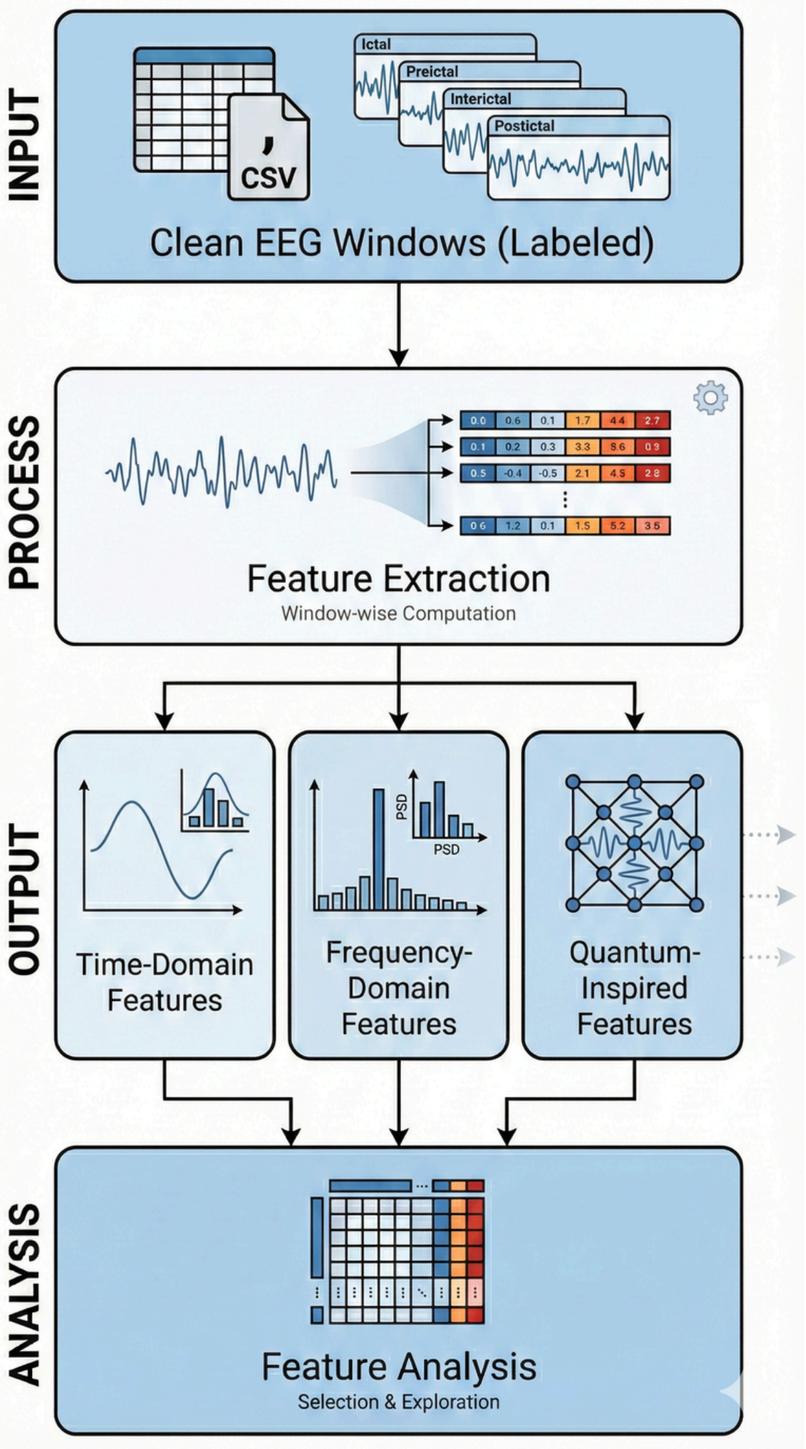
```
34
35 File Name: chb12_06.edf
36 File Start Time: 22:44:34
37 File End Time: 23:44:40
38 Number of Seizures in File: 2
39 Seizure 1 Start Time: 1665 seconds
40 Seizure 1 End Time: 1726 seconds
41 Seizure 2 Start Time: 3415 seconds
42 Seizure 2 End Time: 3447 seconds
43
44 File Name: chb12_08.edf
45 File Start Time: 00:46:38
46 File End Time: 1:46:38
47 Number of Seizures in File: 4
48 Seizure 1 Start Time: 1426 seconds
49 Seizure 1 End Time: 1439 seconds
50 Seizure 2 Start Time: 1591 seconds
51 Seizure 2 End Time: 1614 seconds
52 Seizure 3 Start Time: 1957 seconds
53 Seizure 3 End Time: 1977 seconds
54 Seizure 4 Start Time: 2798 seconds
55 Seizure 4 End Time: 2824 seconds
```

MODULE 1: DATA PRE-PROCESSING

Techniques Used

- **High-pass filtering** – Removes slow baseline drifts and DC offsets by keeping only frequencies above a set threshold (e.g., >0.5 Hz).
- **Notch filtering** – Eliminates narrowband electrical noise, typically 50 Hz or 60 Hz line interference from power sources.
- **Low-pass filtering** – Removes high-frequency noise and muscle artifacts by keeping frequencies below a set cutoff (e.g., <40 Hz).
- **Artifact detection and cleaning** – Identifies and removes or corrects non-brain signals (e.g., eye blinks, muscle activity, movement) to improve EEG signal quality.

MODULE 2
FEATURE EXTRACTION AND ANALYSIS



MODULE 2: FEATURE EXTRACTION AND ANALYSIS

Time Domain Features

Feature Name	Equation	Significance
Hjorth Activity	σ_x^2	Signal variance (or power). A sudden rise often indicates seizure onset.
Hjorth Mobility	$\sqrt{\frac{\text{Var}(x')}{\text{Var}(x)}}$	Estimate of mean frequency. Tracks slowing or fast ripples.
Hjorth Complexity	$\frac{\text{Mobility}(x')}{\text{Mobility}(x)}$	Similarity to a sine wave. Lower values indicate more rhythmic activity.
Zero Crossing Rate	$\frac{1}{T-1}\sum_{t=1}^{T-1}\mathbb{1}(x_t x_{t+1} < 0)$	Frequency proxy. Drastic changes signal transition from chaotic to rhythmic states.
Kurtosis	$\frac{E[(x-\mu)^4]}{\sigma^4}$	Measure of tail heaviness. Detects transient spikes and sharp waves.
Peak-to-Peak Amplitude	$\max(x) - \min(x)$	Dynamic range of the signal. Reflects the intensity of electrical discharges.
Channel Correlation	$\frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$	Spatial synchronization. High values indicate hypersynchrony across brain regions.

MODULE 2: FEATURE EXTRACTION AND ANALYSIS

Frequency Domain Features

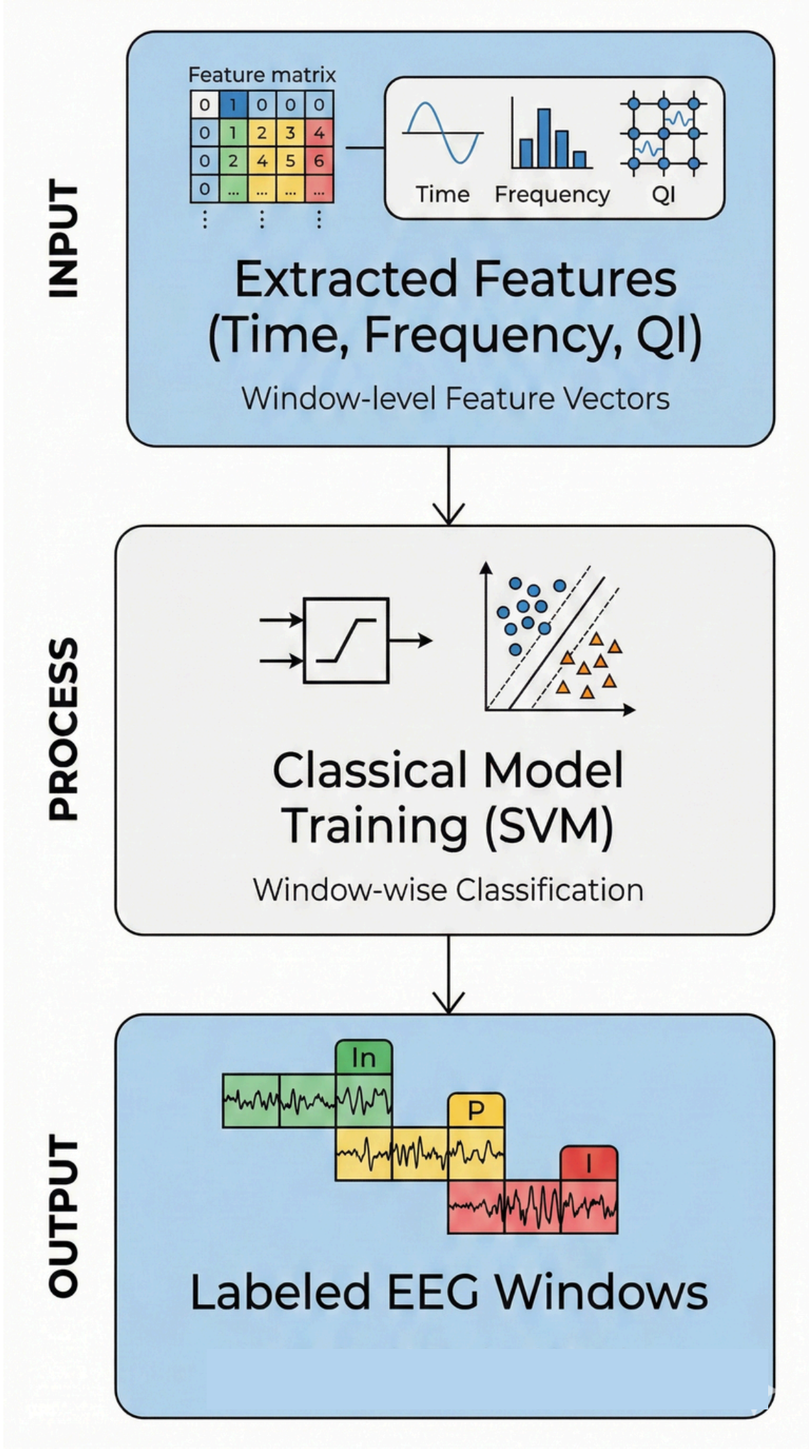
Feature Name	Equation	Significance
Relative Power (Delta)	$\frac{\int_{0.5}^{4} P(f)df}{\int_{0}^{f_{max}} P(f)df}$	Dominance of slow waves. Increased delta often indicates post-ictal slowing.
Relative Power (Gamma)	$\frac{\int_{30}^{80} P(f)df}{\int_{0}^{f_{max}} P(f)df}$	High-frequency content. Gamma oscillations often mark the onset of focal seizures.
Theta/Alpha Ratio	$\frac{P_\theta}{P_\alpha}$	Index of slowing. A shift from healthy alpha rhythm to pathological theta activity.
Spectral Centroid	$\frac{\sum fP(f)}{\sum P(f)}$	Center of mass of the spectrum. Tracks shifts in dominant frequency.
Spectral Entropy	$-\sum \tilde{P}(f) \ln \tilde{P}(f)$	Spectral complexity. Seizures are often highly ordered (low entropy) compared to background.
Spectral Flatness	$\frac{\exp(\frac{1}{N}\sum \ln P_n)}{\frac{1}{N}\sum P_n}$	Tonality measure. Low values (near 0) indicate tonal (rhythmic) oscillations.

MODULE 2: FEATURE EXTRACTION AND ANALYSIS

Quantum Inspired Features

Feature Name	Equation	Significance
Von Neumann Entropy	$-Tr(\rho \ln \rho)$	Quantum information content. Low entropy implies a pure, hypersynchronized state.
Purity	$Tr(\rho^2)$	Measure of state coherence. 1 = Pure state (Seizure), <1 = Mixed state.
Quantum Mutual Info	$S(\rho_A) + S(\rho_B) - S(\rho_{AB})$	Total correlation (quantum + classical). Captures non-linear functional connectivity.
Quantum Fisher Info	$Tr(\rho L^2)$	Sensitivity to parameter change. High values indicate a critical or unstable brain state.
Wigner Entropy	$-\iint W(t,f) \ln W(t,f) dt df$	Complexity in the time-frequency plane. Captures non-stationary seizure dynamics.
Wigner Energy	$\iint \lvert W(t,f) \rvert^2 dt df$	Time-Frequency localization. High energy concentration indicates structured discharges.
Eigenvalue Skewness	$\frac{E[(\lambda - \bar{\lambda})^3]}{\sigma_\lambda^3}$	Asymmetry of covariance eigenvalues. Indicates dominance of specific brain modes.

MODULE 3
Classical Model Development



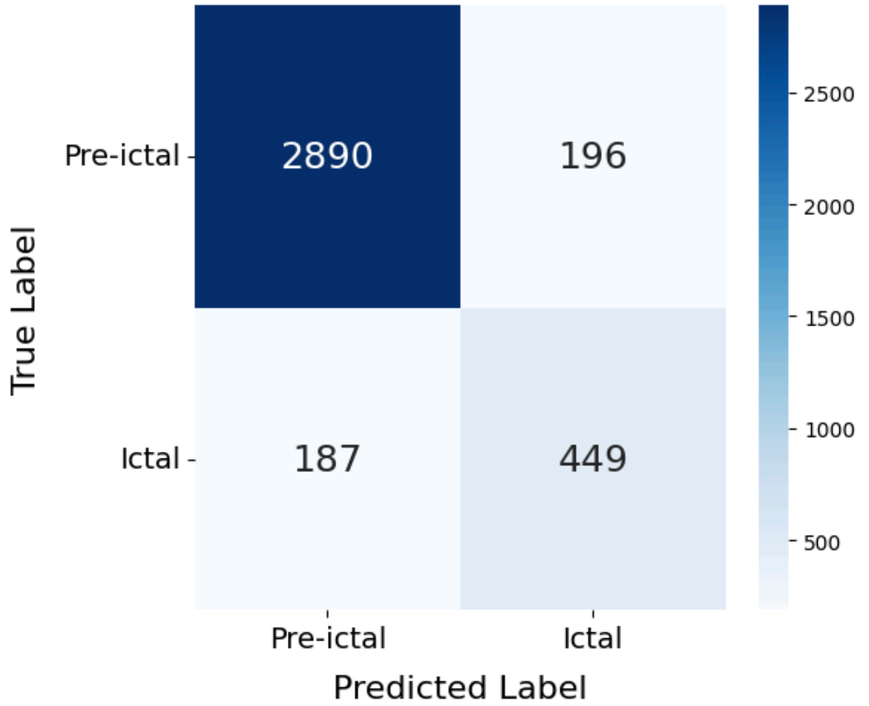
MODULE 3: Classical Model Development

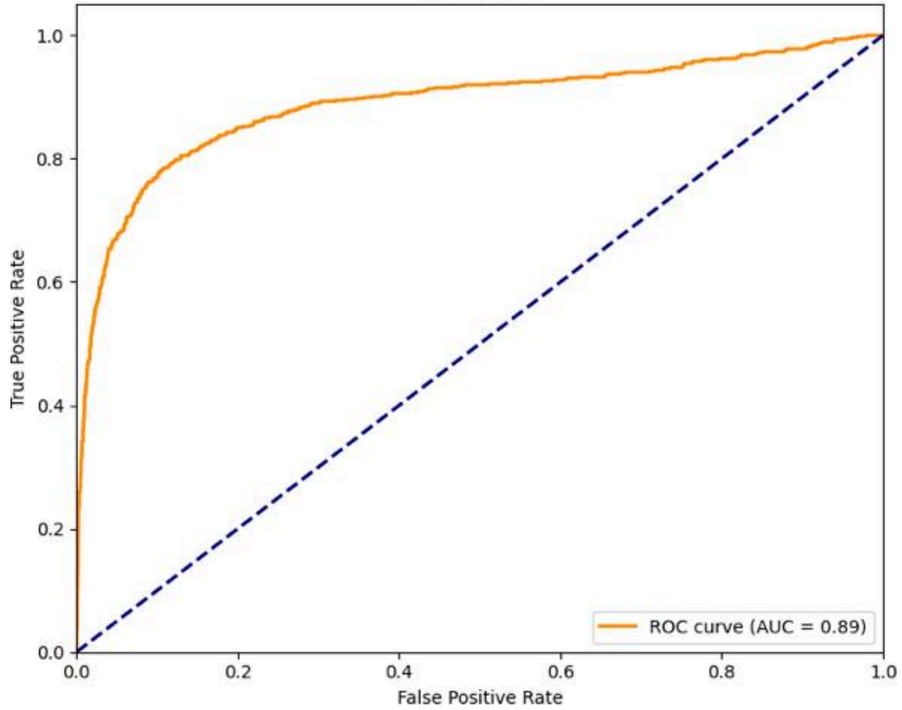
Support Vector Machine (SVM)

- **Hybrid Feature Integration**: Time-domain, frequency-domain, and quantum-inspired features were analyzed both individually and in combination.
- **Patient-Independent Protocol**: Trains on patients 1–20 and Test on patients 21–23.
- **Imbalance Mitigation**: Applies automated class weighting (balanced) to adjust for the disparity between preictal and ictal sample counts.

MODULE 3: Classical Model Development

Time Domain Features Only

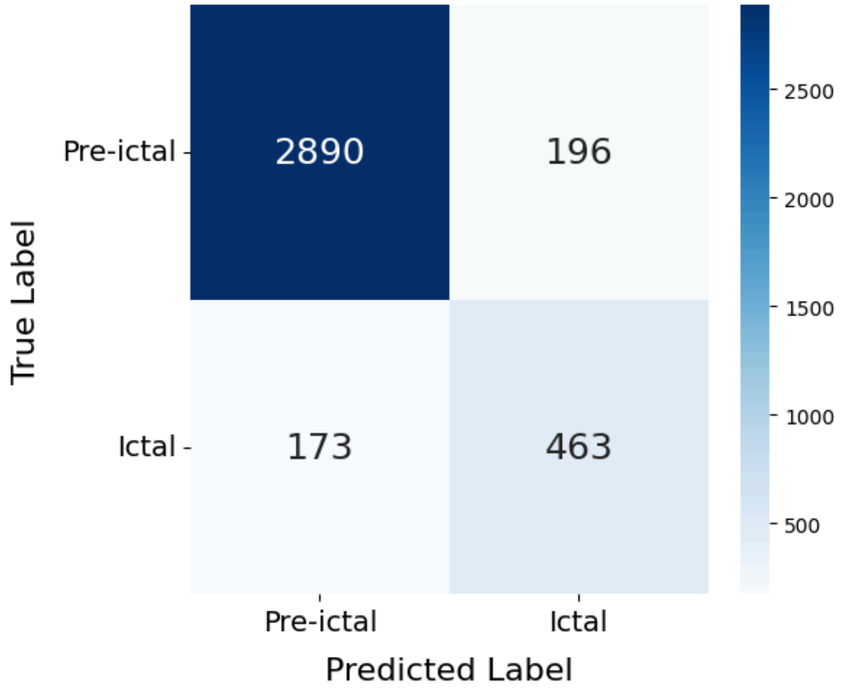


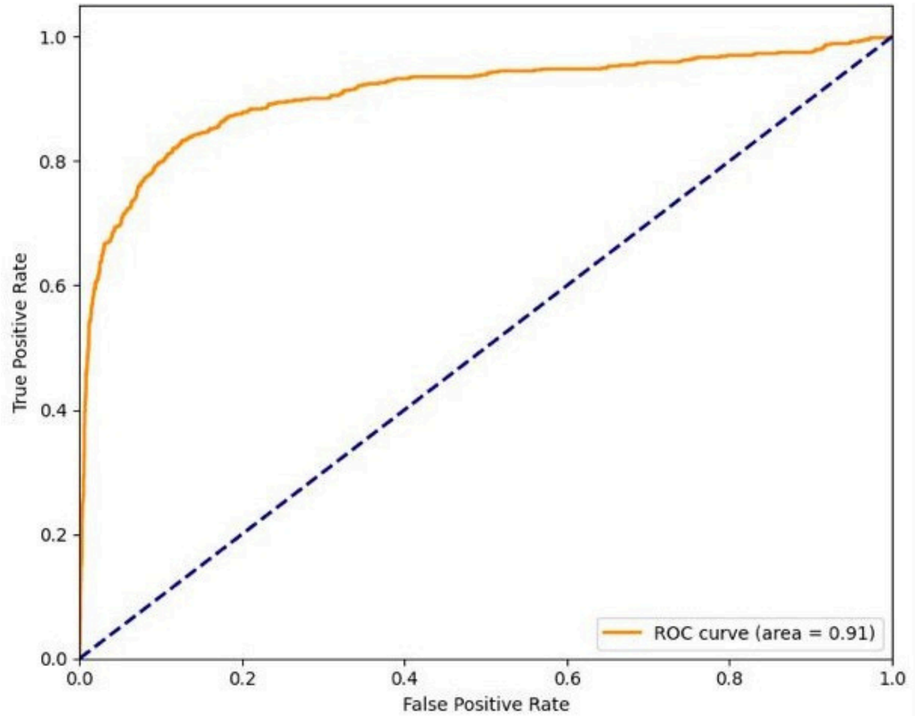


	Precision	Recall	ROC-AUC
preictal	0.94	0.94	0.889
ictal	0.7	0.71	

MODULE 3: Classical Model Development

Frequency Domain Features Only

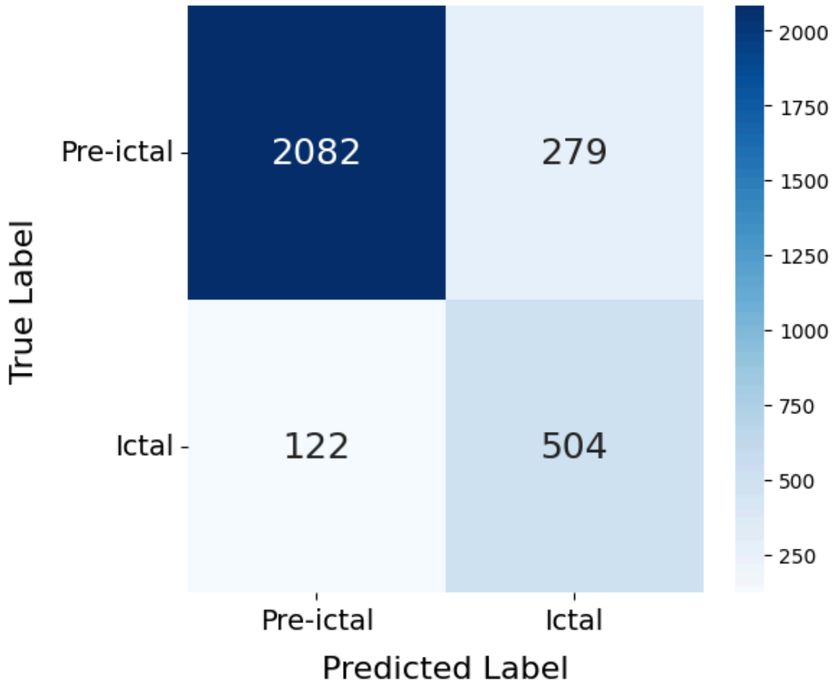


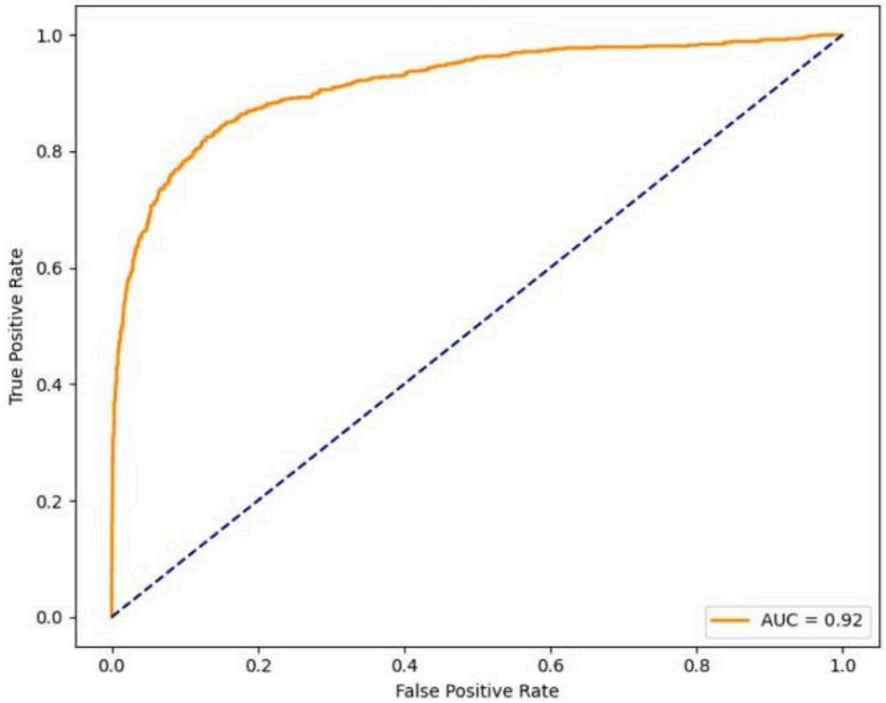


	Precision	Recall	ROC-AUC
preictal	0.94	0.94	0.9078
ictal	0.7	0.73	

MODULE 3: Classical Model Development

Time-Frequency Domain Features Only

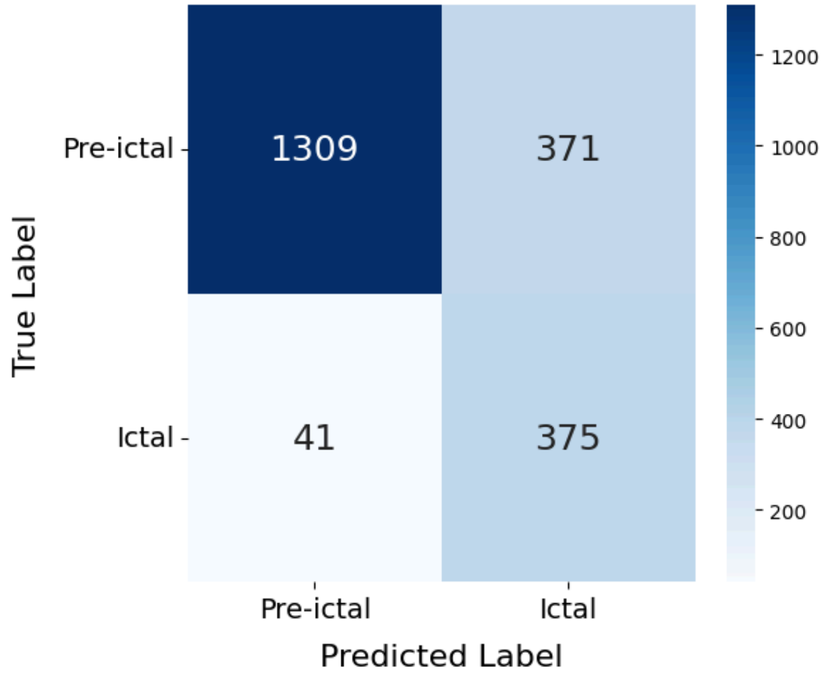


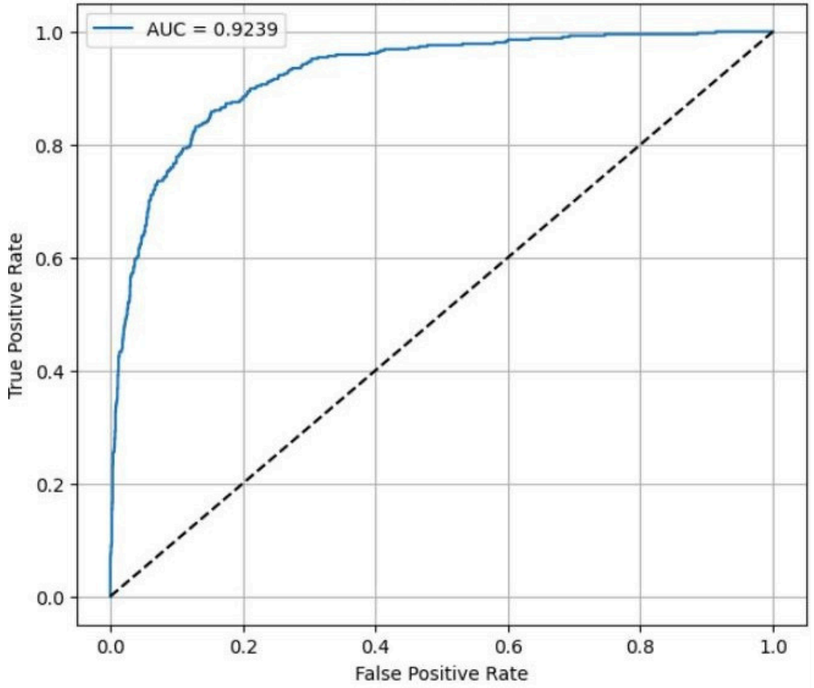


	Precision	Recall	ROC-AUC
preictal	0.94	0.88	0.9151
ictal	0.64	0.81	

MODULE 3: Classical Model Development

Quantum Inspired Features Only

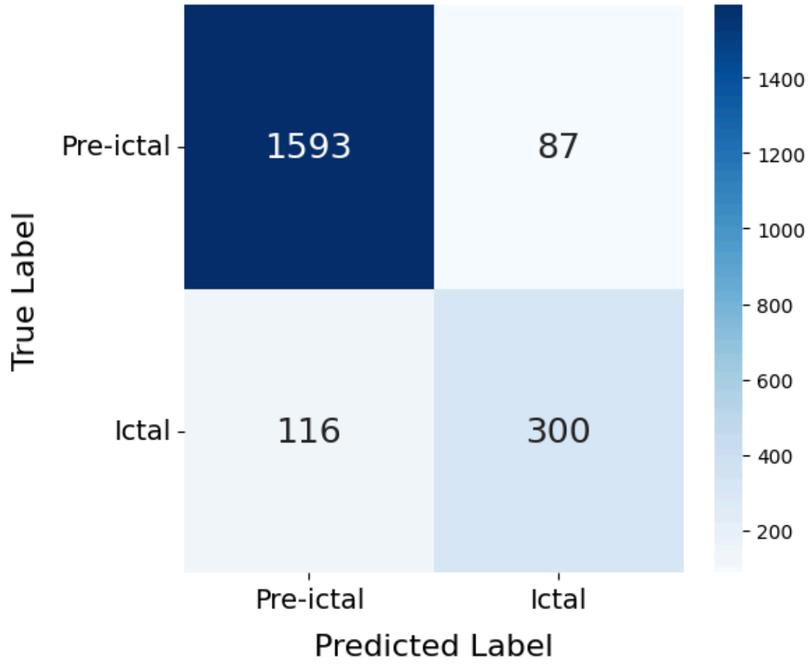


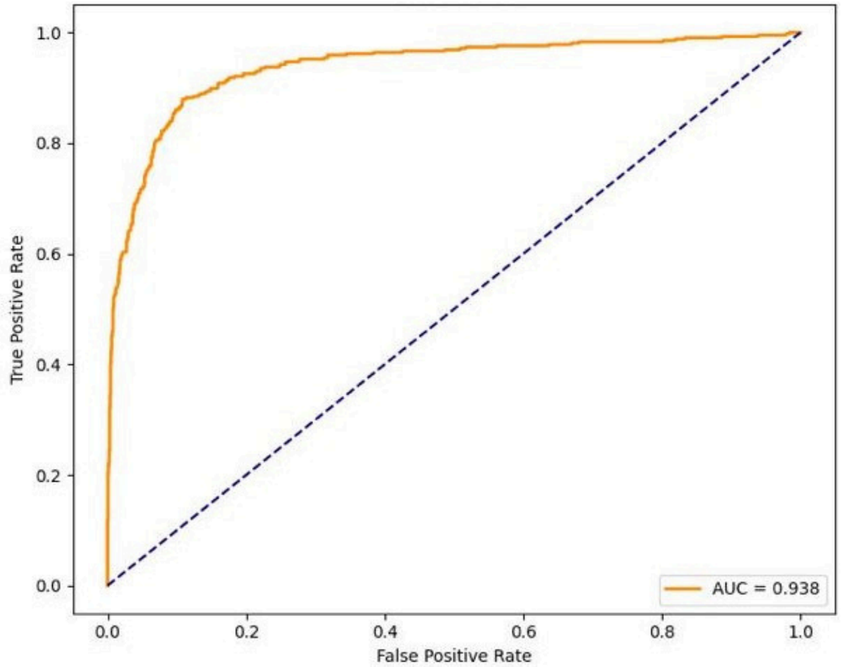


	Precision	Recall	ROC-AUC
preictal	0.97	0.78	0.9239
ictal	0.5	0.9	

MODULE 3: Classical Model Development

Time-Frequency-Quantum Inspired Features Combined



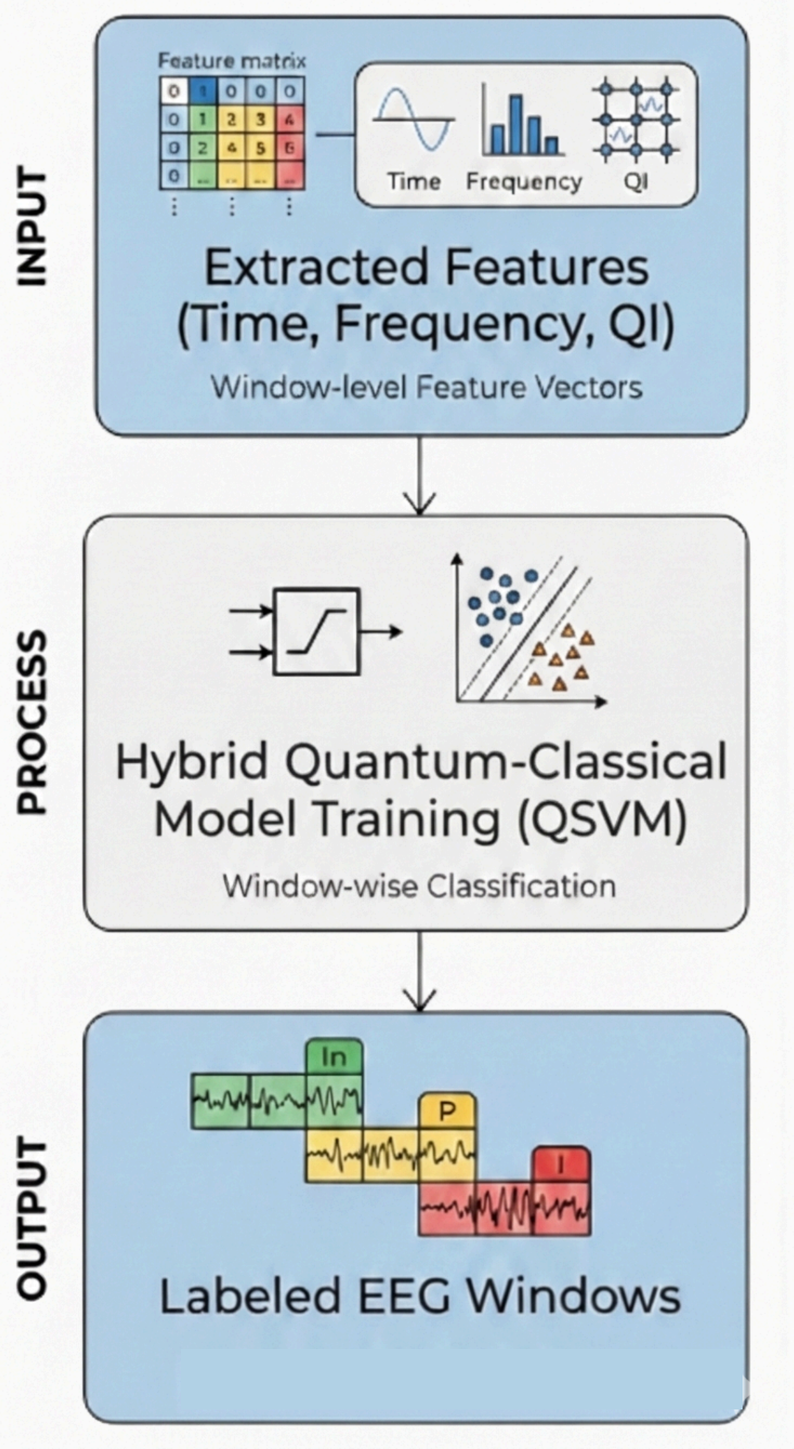


	Precision	Recall	ROC-AUC
preictal	0.93	0.95	0.938
ictal	0.78	0.72	

MODULE 3: Classical Model Development

Class	Precision	Recall	F1-Score
Pre-ictal (0)	0.93	0.95	0.94
Ictal (1)	0.78	0.72	0.75
Overall Accuracy: 0.90			

MODULE 4
Hybrid Quantum-Classical Model Development



MODULE 4: Hybrid Quantum Classical Model Development

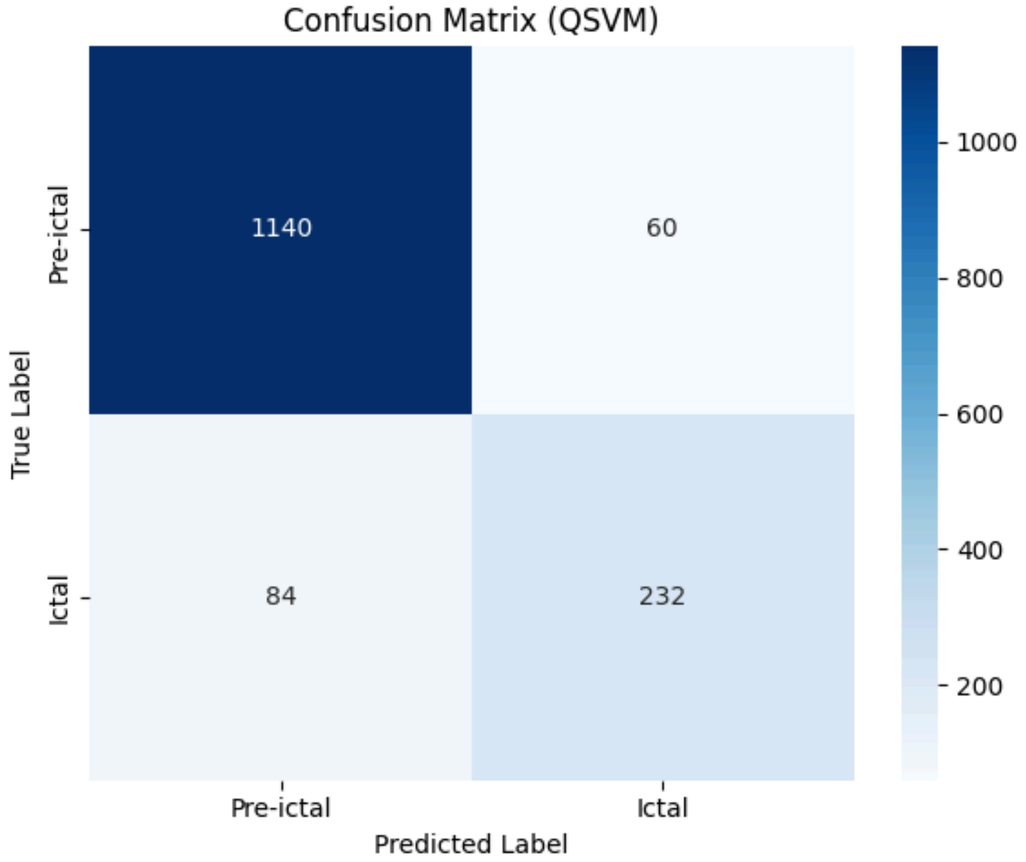
Quantum Support Vector Machine

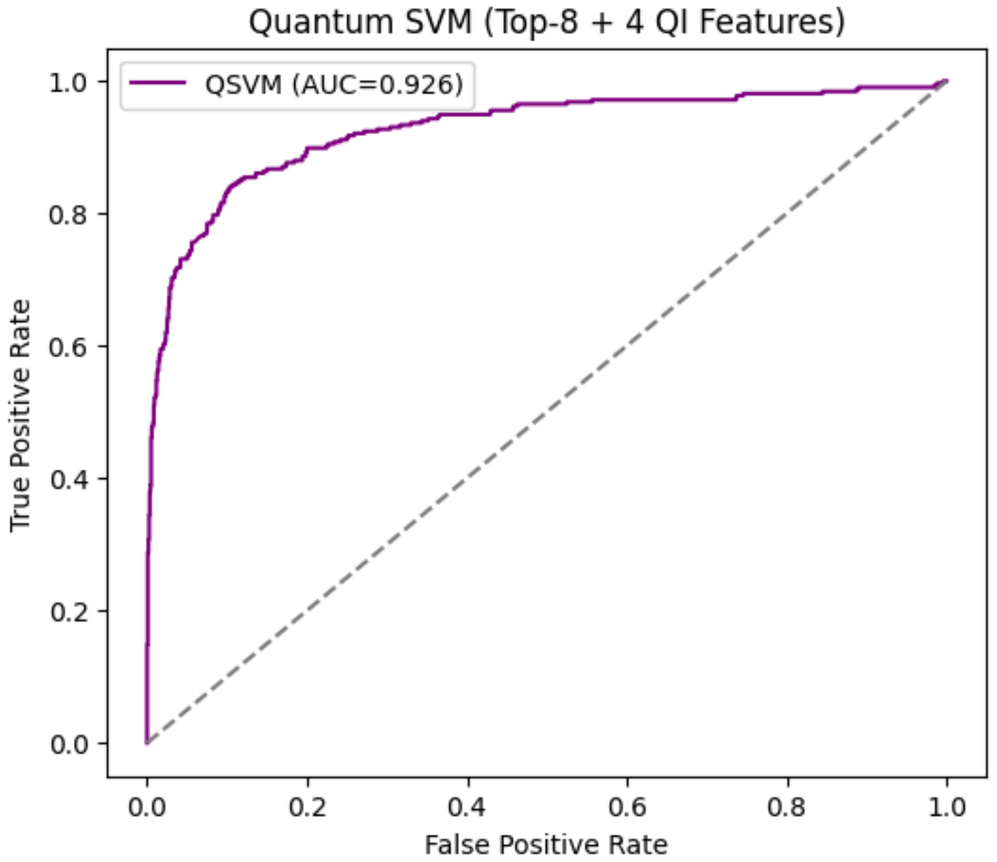
- **Hybrid Feature Selection**: Top 8 features selected using ANOVA F-test + 4 Quantum-Inspired features
- **Patient-Independent Protocol**: Trained on patients 01–20, tested on unseen patients 21–23 for clinical generalization.
- **Imbalance Mitigation**: Per-patient balancing and global class equalization
- **Hybrid Learning Framework:** Quantum kernel computation and classical SVM optimization

MODULE 4: Hybrid Quantum-Classical Model Development

Feature Name	Equation	Seizure Relevance
Von Neumann Entropy	$-\sum \lambda_i \log \lambda_i$	Quantum information content; decreases with hypersynchronization.
Entanglement Entropy (Half)	$S(\rho_{sub})$	Inter-channel coupling strength; elevated during ictal states.
Eigenvalue Skewness	$Skew(\lambda)$	Asymmetry in network spectrum; reflects structural shifts.
Quantum Fisher Information	$Var(\lambda)$	Sensitivity to perturbation; captures instability.
Spectral Flatness	GM/AM	Tonality vs noise; seizures reduce flatness.
Global Std Dev	$\sigma(x)$	Overall amplitude variability.
Mean Absolute Amplitude	$Mean(\|x\|)$	Average signal magnitude.
Median Absolute Amplitude	$Median(\|x\|)$	Robust signal intensity measure.
Amplitude Range	$\max - \min$	Extreme excursions; spike sensitivity.
Mean Channel Std	$Mean(\sigma_c)$	Spatial amplitude dispersion.
Mean Peak-to-Peak	$Mean(P2P)$	Captures sharp epileptic discharges.
Time-Domain Std	σ_{global}	Overall time-domain variability.

MODULE 4: Hybrid Quantum-Classical Model Development





MODULE 4: Hybrid Quantum-Classical Model Development

Class	Precision	Recall	F1-Score
Pre-ictal (0)	0.93	0.95	0.94
Ictal (1)	0.79	0.73	0.76
Overall Accuracy: 0.91			

MODULE 5: Model Evaluation and Comparison

All Standard Evaluations Techniques will be used like:

- Accuracy
- Precision, Recall, F1-score
- Confusion Matrix
- ROC–AUC

Quantum Advantage / Quantum Gain

Measures the performance improvement of the quantum component over a purely classical approach.

$$Q_{\text{gain}} = \frac{\text{Metric}_{\text{quantum}} - \text{Metric}_{\text{classical}}}{\text{Metric}_{\text{classical}}}$$

MODULE 5: Model Evaluation and Comparison

- Accuracy improved by ~1.1% (0.90 → 0.91).
- Ictal Precision improved by ~1.3% (0.78 → 0.79).
- Ictal Recall improved by ~1.4% (0.72 → 0.73).
- Ictal F1-score improved by ~1.3% (0.75 → 0.76).

- The hybrid quantum–classical model achieved comparable performance using only 12 features, compared to 56 features required by the classical model.

- However, the quantum model introduces higher computational cost due to quantum kernel evaluation, making it more computationally intensive.

Epileptic Seizure State Classification Using Quantum Inspired Techniques

Sminu Izudheen[1], Franklin Davis Achandy[1*], Joyal George Joseph[1], Colin Pradeep[1], Jose Chemmanoor[1]

[1]Department of Computer Science and Engineering, Rajagiri School of Engineering & Technology, Kochi, India .

*Corresponding author(s). E-mail(s): **fda10h04@gmail.com**;

Abstract

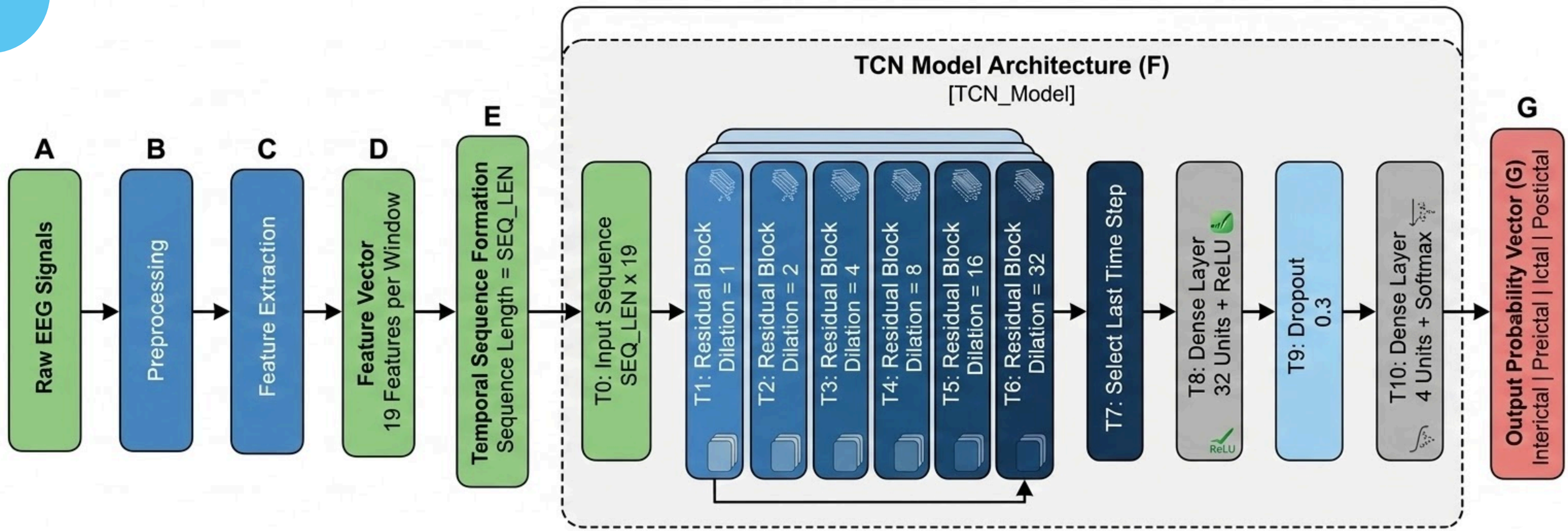
Accurate discrimination between preictal and ictal states from electroencephalography (EEG) signals is essential for epilepsy management and represents a key step toward reliable seizure monitoring and early warning systems. However, the complex, non-stationary, and highly patient-specific nature of EEG signals continues to limit the discriminative capability of conventional feature representations. This study investigates whether quantum-inspired (QI) descriptors can provide a more informative and robust feature space for epileptic seizure state classification. Seizure state discrimination is evaluated across three feature domains: time-domain features, frequency-domain features, and quantum-inspired descriptors designed to capture global signal structure and energy redistribution characteristics. Classification is performed using a support vector machine (SVM) with an RBF kernel under a strict patient-independent evaluation protocol on the CHB-MIT Scalp EEG dataset, enabling assessment of generalization to unseen subjects. Experimental results demonstrate that quantum-inspired features achieve stronger discriminative performance than individual time-domain or frequency-domain representations, and provide performance comparable to their classical combination. Furthermore, integrating time-domain, frequency-domain, and quantum-inspired features yields the best overall performance, achieving an ROC–AUC of 0.9376 on unseen patients, with overall accuracy remaining consistently high across configurations. These findings suggest that quantum-inspired feature representations capture complementary system-level information that enhances seizure state classification when combined with conventional descriptors.

Keywords: Epileptic seizure classification, Electroencephalography, Quantum-inspired features, Patient-independent evaluation, Support vector machine

Towards an Early Warning System

- Early seizure detection is more valuable than abnormal state classification.
- It enables timely intervention and improves patient safety.
- We extend our work to detect transitions leading to seizure onset.
- A Temporal Convolutional Network (TCN) is used to model these temporal patterns.

Architecture



WHY TCN?

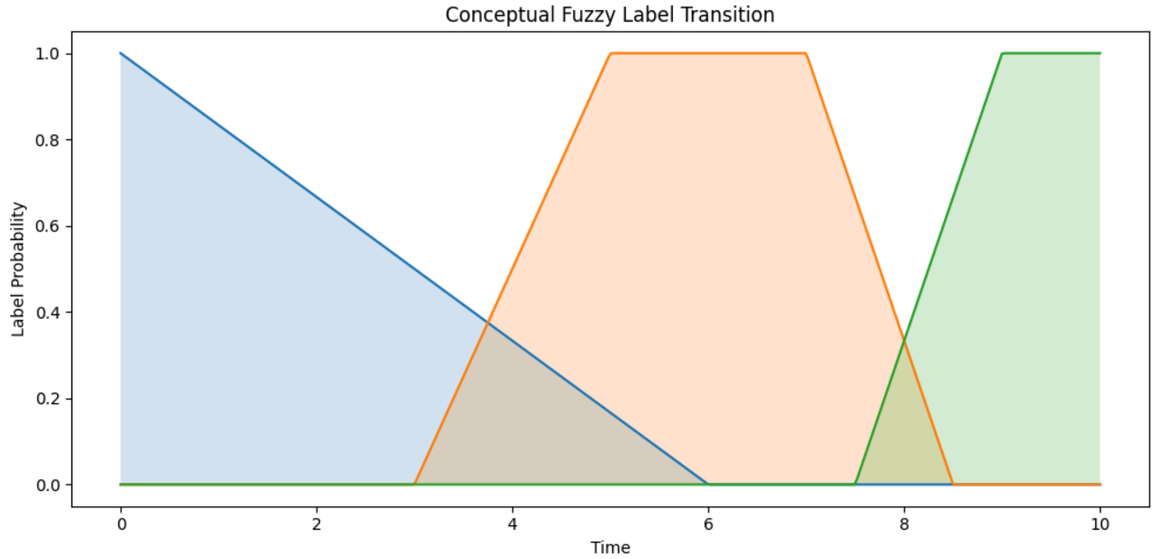
- **CNN** capture local patterns in feature representations but are limited in modeling long-term temporal dependencies.

- **LSTM** model temporal data but rely on sequential processing and may struggle with capturing very long-range dependencies effectively.

- **TCN** use convolutional operations to model temporal dependencies, enabling parallel computation and effective capture of long-range patterns.

Module 1 : Data preprocessing

- Bandpass and notch filters are applied to remove baseline drift, high-frequency noise, and power-line interference.
- EEG signals are segmented into 4-second windows with 50% overlap
- Fuzzy (soft) labeling is applied near seizure boundaries to represent gradual transitions between states



Module 2 : Feature Extraction

- EEG windows are transformed into a set of 19 quantitative features representing spectral characteristics, signal dynamics, and global system behaviour.

Feature	Equation	Interpretation
Hjorth Activity	$Var(x)$	Overall signal power or energy level
Hjorth Mobility	$\sqrt{\frac{Var(dx)}{Var(x)}}$	Average frequency content of the signal
Hjorth Complexity	$\frac{\sqrt{Var(ddx)/Var(dx)}}{Mobility}$	Irregularity and waveform complexity
Higuchi FD (k=6)	$HFD(x, 6)$	Short-scale fractal complexity of EEG signal
Higuchi FD (k=50)	$HFD(x, 50)$	Long-scale fractal complexity of EEG signal
Wigner Skewness	$skew(\lvert H(x)\rvert)$	Asymmetry of signal envelope distribution
Von Neumann Entropy	$-\sum \lambda_i \log(\lambda_i)$	Global disorder of multichannel covariance structure
Entanglement Entropy	$-\sum \lambda_i^{(L)} \log(\lambda_i^{(L)})$	Interaction strength between channel groups
Quantum Fisher Information	$\sum(\rho_t - \rho_{t-1})^2$	Sensitivity to temporal changes in system state
Energy Change	$\lvert E(t) - E(t-1)\rvert$	Magnitude of change in global signal energy

Module 2 : Feature Extraction

Feature	Equation	Interpretation
Theta / Delta Ratio	θ/δ	Balance between slow (delta) and mid-frequency activity
Alpha / Theta Ratio	α/θ	Relationship between alpha rhythm and slower theta activity
Beta / Alpha Ratio	β/α	Shift toward faster oscillatory components
Gamma / Beta Ratio	γ/β	Relative presence of high-frequency activity
Delta / Alpha Ratio	δ/α	Slow-wave dominance compared to alpha rhythm
Delta / Beta Ratio	δ/β	Slow activity compared to faster beta rhythm
Theta / Beta Ratio	θ/β	Interaction between theta and beta rhythms
Relative Delta Power	$\delta/(\delta+\theta+\alpha+\beta+\gamma)$	Proportion of slow-wave power in total spectrum
Relative Theta Power	$\theta/(\delta+\theta+\alpha+\beta+\gamma)$	Relative contribution of theta band activity

Module 3 : Model Development

- **Input**: Sequential EEG feature vectors representing temporal evolution across windows
- **Model**: Temporal Convolutional Network (causal sequence modeling)
- **Temporal Learning**: Captures long-range dependencies and preictal transition patterns
- **Output**: Probability distribution over seizure stages (interictal, preictal, ictal, postictal)

Module 3 : Model Development

- **Smoothing**: Temporal averaging across consecutive windows to reduce prediction noise
- **State Mapping**: Grouping of stage probabilities to emphasize preictal and ictal detection
- **Decision Logic:** Threshold-based identification of state transitions for early warning

Results

CHB08

Seizure Event	True Seizure Time (s) - A	Ictal Detected Time (s)	Preictal Warning Time (s) - B	Preictal Lead Time (s) A-B
Seizure 1 (Training)	2670	2680	278	2392
Seizure 2 (Training)	2856	2868	2040	816
Seizure 3 (Validation)	2988	3002	2724	264
Seizure 4 (Unseen)	2417	2430	898	1519
Seizure 5 (Unseen)	2083	2096	1646	437

- An average preictal lead time of ~1086 seconds (~18 minutes) highlights the model's capability for meaningful early seizure warning.

Results

CHB10

Seizure Event	True Seizure Time (s) - A	Ictal Detected Time (s)	Preictal Warning Time (s) - B	Preictal Lead Time (s) A-B
Seizure 1 (Training)	6313	6314	5212	1101
Seizure 2 (Training)	6888	6860	5942	946
Seizure 3 (Validation)	2382	2384	2112	270
Seizure 4 (Unseen)	3021	3040	2146	875
Seizure 5 (Unseen)	3801	3818	3008	793
Seizure 6 (Unseen)	4618	4634	238	4380
Seizure 7 (Unseen)	1383	1388	238	1145

- An average preictal lead time of ~1358 seconds (~22 minutes) highlights the model's capability for meaningful early seizure warning.

Feature Ablation Study

CHB08

Cluster	Grouped Features	Sens (%)	FPR/h	Lead Time (min)	Missed	Alarms
Baseline	–	100.0	0.62	24.73	0	17
Cluster_6	Theta/Delta, Relative Theta Power	100.0	0.46	26.72	0	15
Cluster_4	Alpha/Theta	100.0	0.92	24.87	0	20
Cluster_1	Beta/Alpha, Gamma/Beta, HFD (k=6)	100.0	0.62	24.63	0	16
Cluster_7	Delta/Alpha, Relative Delta Power	100.0	0.69	26.70	0	17
Cluster_2	Delta/Beta, Theta/Beta, Hjorth Mobility, HFD (k=50)	50.0	0.23	36.15	1	7
Cluster_8	Hjorth Activity	100.0	0.69	24.70	0	15
Cluster_3	Hjorth Complexity	50.0	0.15	36.15	1	6
Cluster_10	Wigner Skewness	100.0	0.62	24.63	0	15
Cluster_5	Von Neumann Entropy, Entanglement Entropy	100.0	0.85	24.73	0	21
Cluster_11	QFI (Quantum Fisher Information)	100.0	0.62	24.70	0	16
Cluster_9	Energy Change	100.0	0.54	24.65	0	16

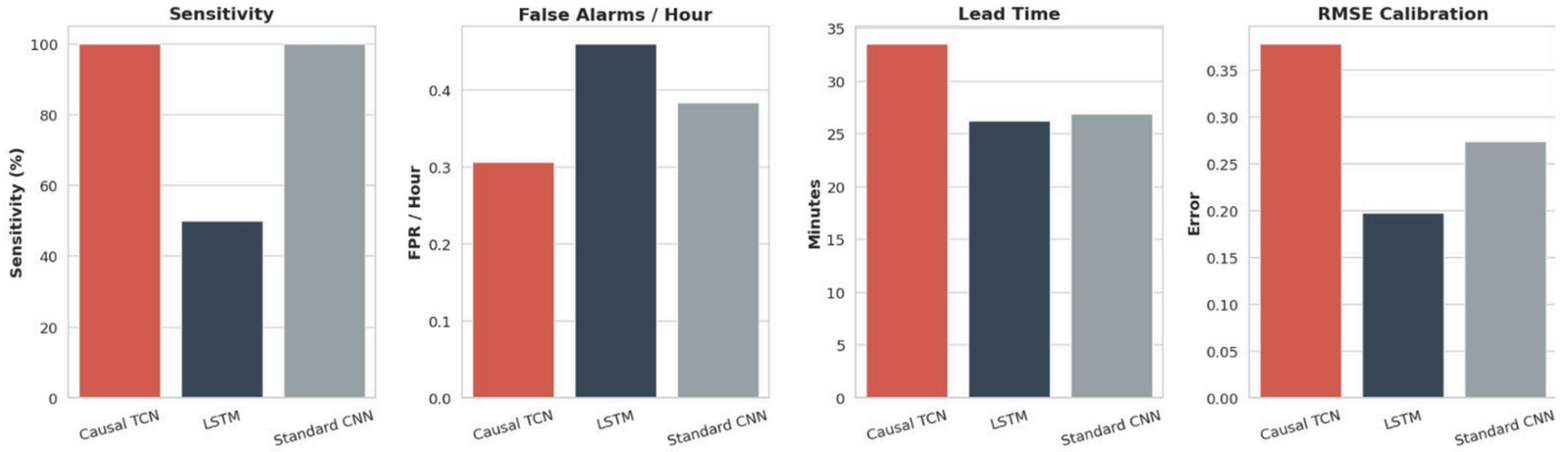
CHB10

Cluster	Grouped Features	Sens (%)	FPR/h	Lead Time (min)	Missed	Alarms
Baseline	–	100.0	1.0	32.24	0	15
Cluster_10	Theta/Delta, Relative Delta Power, Relative Theta Power	75.0	0.6	30.47	1	9
Cluster_6	Alpha/Theta, Theta/Beta	100.0	0.9	32.35	0	14
Cluster_4	Beta/Alpha, Higuchi FD (k=6)	100.0	1.1	32.27	0	16
Cluster_5	Gamma/Beta	100.0	1.0	36.16	0	16
Cluster_8	Delta/Alpha	100.0	1.1	32.34	0	16
Cluster_7	Delta/Beta, Hjorth Mobility, Higuchi FD (k=50)	100.0	0.8	36.18	0	14
Cluster_1	Hjorth Activity	100.0	1.0	32.35	0	15
Cluster_11	Hjorth Complexity	100.0	1.0	32.30	0	15
Cluster_9	Wigner Skewness	100.0	1.2	36.12	0	18
Cluster_3	Von Neumann Entropy, Entanglement Entropy	100.0	1.2	41.48	0	19
Cluster_12	QFI Proxy	100.0	1.2	32.31	0	17
Cluster_2	Energy Change	100.0	0.9	32.24	0	14

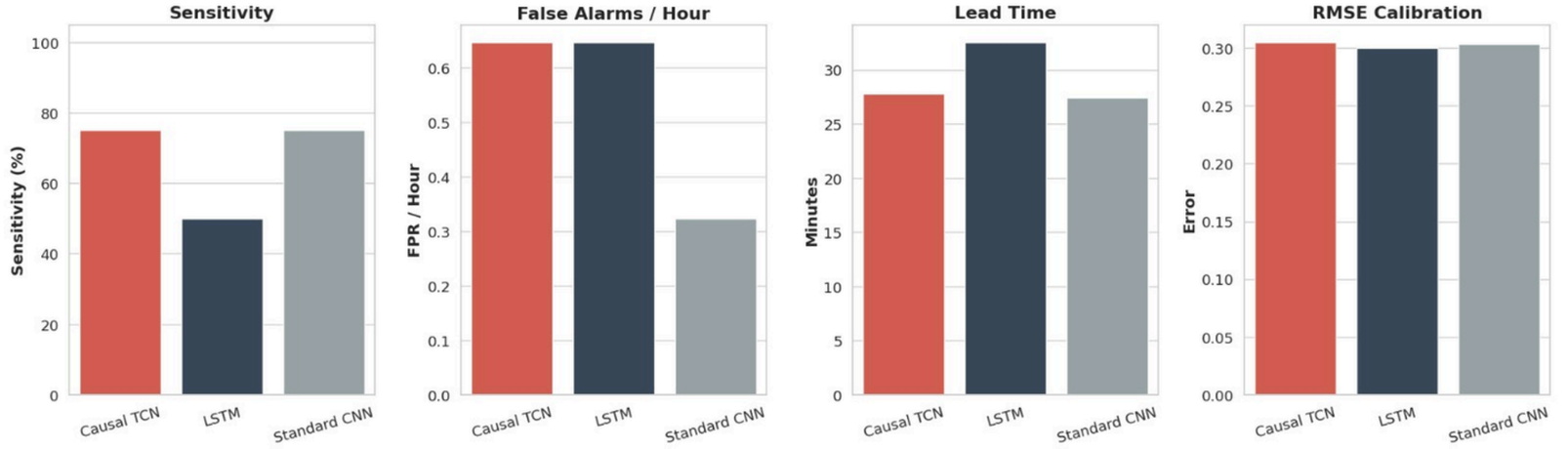
- Variation in performance across ablations suggests that different features contribute differently for each patient, highlighting patient-specific EEG dynamics.

Model Comparison

CHB08

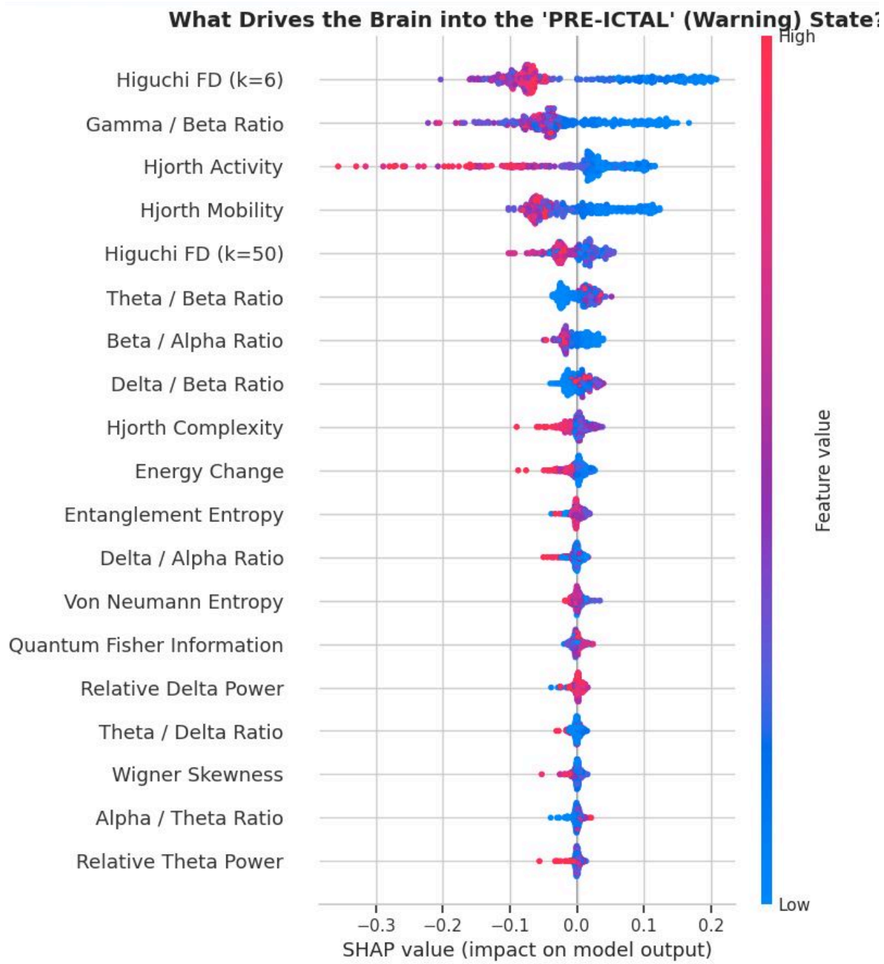


CHB10



- TCN provides the best overall trade-off between sensitivity, false alarms, and lead time, making it the most balanced model across both patients.

Feature Contributions (SHAP)





- SHAP analysis highlights that both classical and quantum-inspired features significantly influence preictal prediction, with their importance varying across patients.

LIMITATIONS

- The current system uses a patient-specific model, requiring training on historical EEG data for each individual.
- The approach relies on hand-crafted feature extraction, which may not capture all relevant patterns present in raw EEG signals.
- EEG recordings often contain noise, artifacts, and variability, which can affect feature stability and model performance.
- The system has been evaluated on a limited dataset and requires further validation on larger and more diverse patient populations.

HARDWARE & SOFTWARE REQUIREMENTS

- Software Requirements
  - Python
  - Libraries: TensorFlow, Scikit-learn, NumPy, SciPy, Pandas
  - EEG Processing: MNE, PyEDFlib
  - Quantum Framework: PennyLane

- Hardware Requirements
  - GPU-enabled environment (Google Colab)
  - Minimum 12 GB RAM recommended
  - Multi-core CPU for signal processing
  - Storage for EEG datasets and trained models

CONCLUSION

- This study explored the analysis of EEG signals across different seizure stages using feature-based modeling approaches.
- Various time-domain, frequency-domain, and quantum-inspired features were investigated, along with the potential of quantum–classical hybrid models for analyzing complex EEG dynamics.
- A TCN-based temporal framework was developed to capture the evolution of brain activity over time and detect transitions between seizure stages.
- Future work could focus on improving robustness to noise and artifacts, incorporating multimodal physiological data, and validating the framework on larger and more diverse datasets.

Patient-Specific Preictal State Identification Using Hybrid Classical and Quantum-Inspired Feature Learning

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***Abstract*—This study focuses on patient-specific identification of the preictal phase in epileptic EEG signals. Accurate recognition of the preictal state is clinically important for enabling proactive therapeutic intervention before seizure onset. The proposed framework integrates classical time-domain and frequency-domain EEG features with quantum-inspired system-level representations to capture both localized and global multichannel dynamics.**

***Index Terms*—Epileptic seizure prediction, Preictal identification, Patient-specific modeling, EEG analysis**

I. Introduction

Epilepsy is a chronic neurological disorder characterized by recurrent and unprovoked seizures resulting from abnormal and excessive neuronal synchronization in the brain [1]. According to the World Health Organization, epilepsy affects more than 50 million individuals worldwide, imposing substantial clinical, social, and economic burden [2].

A seizure is not an instantaneous event but a dynamic process that evolves through distinct physiological phases. These phases are typically categorized as: (i) the *interictal* phase, representing baseline brain activity between seizures; (ii) the *preictal* phase, denoting the transitional period preceding seizure onset; (iii) the *ictal* phase, corresponding to the seizure event itself; and (iv) the *postictal* phase, reflecting recovery following seizure termination [3]. Understanding and distinguishing these phases is fundamental to developing effective seizure monitoring and intervention systems.

Most existing automated systems focus on ictal detection, which identifies seizure occurrence after clinical onset [4]. While ictal detection is valuable for retrospective confirmation and monitoring, it does not provide sufficient lead time for preventive intervention. In contrast, identification of the preictal phase aims to recognize early electrophysiological alterations that precede seizure manifestation [3]. Accurate preictal identification enables proactive therapeutic strategies such as adaptive neurostimulation and real-time alert systems, thereby significantly improving patient safety and autonomy [5].

However, reliable preictal identification is considerably more challenging than ictal detection. The transition from interictal to ictal states is gradual and often subtle, characterized by complex, non-stationary, and patient-specific EEG dynamics [3]. Inter-subject variability in seizure origin, cortical organization, electrode configuration, and physiological conditions further limits the generalizability of population-level models [6]. These challenges motivate the development of patient-specific frameworks explicitly designed to model preictal dynamics.

Although seizure prediction has been extensively investigated over the past two decades, many studies formulate the problem as binary seizure versus non-seizure discrimination or ictal onset detection [7], [8]. Comparatively fewer studies explicitly formulate preictal state identification as a distinct modeling objective [9]. Given its critical role in anticipatory intervention, preictal state identification warrants focused methodological development.

In this study, we propose a patient-specific framework for preictal state identification that integrates classical EEG descriptors with quantum-inspired system-level representations. By combining localized statistical measures with global inter-channel modeling, the proposed approach aims to enhance characterization of subtle preictal transition patterns while maintaining interpretability and computational feasibility.

II. Related Work

A. Seizure Detection and General Prediction Frameworks

Early research in automated seizure analysis primarily addressed ictal detection using handcrafted time-domain and frequency-domain EEG features, including signal variance, spectral power, energy measures, and Hjorth parameters [4]. Spectral estimation techniques such as Welch's method have been extensively employed to quantify band-specific energy redistribution during seizure activity.

Comprehensive reviews of seizure prediction research have highlighted the role of nonlinear dynamics, synchronization indices, and complexity measures in characterizing seizure evolution [3], [5]. Subsequent approaches incorporated entropy-based metrics and correlation-based features to improve discrimination between seizure and non-seizure states.

More recently, deep learning architectures such as convolutional neural networks (CNNs) and long short-term memory (LSTM) networks have demonstrated promising performance in seizure detection and prediction tasks [7], [8]. Nevertheless, a substantial portion of these studies focus on seizure versus

ONGOING & FUTURE WORKS

- This work is being extended into a funded research project under the AICTE Research Promotion Scheme (RPS).

- **Project Title**: NeuroFusion: Edge-Deployable AI Platform for Multimodal Neurological Monitoring
- **Focus**: Development of an integrated, real-time system combining multimodal data for neurological monitoring
- **Grant Amount**: ₹ 20,48,000
- **Co-PI**: Dr. Sminu Izudheen

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THANK YOU